Yunqi Huang

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EDUCATION

Ph.D. in Computer Science, **University College London**, London, UK 2023 - Present

Supervisor: Prof. Thomas George Thuruthel

My research focuses on robot learning, tactile perception, and learning-based control for soft robotic systems, with the goal of enabling robots to learn robust and adaptive behaviours through multimodal sensing and physical interaction. Experienced in the design, fabrication, and characterisation of soft robots and tactile sensors, together with signal processing, tactile image reconstruction, sensor fusion, and deep learning.

Technical Skills: Python, MATLAB, ROS2, Arduino, Solidworks, Rapid Prototyping (3D Printing, Silicone Casting)

Awards: UCL Research Scholarship (Fully Funded)

Master of Engineering in Robotics, **National University of Singapore**, Singapore 2021 - 2023

Supervisor: Prof. Chui Chee Kong

M.Eng Project: Ocular Surgical Robot with a Novel Vacuum Actuator for Vitreoretinal Surgery

- Designed an RCM (Remote Center of Motion) mechanism robotic arm and a novel soft vacuum actuator to assist surgeons in holding surgical tools without hand tremors.
- Implemented control algorithms and reinforcement learning to enable cooperative movement between the robotic arm and the surgeon's hand.

Awards: NUS GRIP, First Prize at Huawei Tech4City Competition 2022

Bachelor of Science in Mechanical Engineering, **Harbin Institute of Technology**, China 2016 - 2020

Supervisor: Prof. Jingfeng He

B.Sc Project: Vibration Actuated Spherical Tensegrity Robot Locomotion

- Designed a spherical tensegrity robot for exploration, including structure design, robot fabrication, dynamic model simulation and control of the robot's vibration-driven gaits.

Awards: People's Scholarship in China, National First Prize in 2018 China RoboWork Competition

RESEARCH EXPERIENCE

Visiting Researcher, **Soft Robotics Lab**, **ETH Zurich**, Switzerland July 2026 - September 2026

Host: Prof. Robert Katzschmann

- Designed and fabricated a palm-shaped tactile skin for the ORCA Hand, an open-source anthropomorphic robotic hand. Trained a tactile-driven grasping policy that closes the control loop on contact feedback.

Research Assistant, **The Chinese University of Hong Kong**, Hong Kong SAR May 2023 - September 2023

Host: Prof. Qi Dou

- Developed an adaptive haptic-assisted training framework for surgical robot simulators using reinforcement learning and shared control methods. Integrated RL-predicted trajectories with haptic feedback to enhance surgical skills training in a simulated environment.

Research Assistant, **Harbin Institute of Technology (Shenzhen)**, China September 2020 - July 2021

Host: Prof. Honghai Liu

- Developed interactive intervention games on Ubtech Cruzr for children with autism spectrum disorder (ASD), translating clinical protocols into real-time robot behaviours.

LISTED PUBLICATIONS

According to the Google Scholar Database as of September 2026: H-index 5, the total number of citations is 68, and 11 documents. **Full list available here**

[1] **Huang, Yunqi**, Q. Lin, and T. G. Thuruthel. "Investigation on Non-Uniform Conductivity Distribution in EIT-Based Tactile Skins for Enhanced Localized Sensitivity". *IROS*. 2026.

[2] K. Nazari, **Huang, Yunqi**, D. Hardman, F. Iida, and T. G. Thuruthel. "Touch with Insight: Physics-Aware Data-Driven Learning for EIT-Based Tactile Sensing". *ICRA*. 2026. DOI: 10.1109/ICRA57385.2026.11696850.

[3] **Huang, Yunqi**, G. Pei, J. Hughes, and T. G. Thuruthel. "Visual Proprioception and Contact Prediction for Soft Robotic Arms Using Onboard Sensing and Generative Models". *IEEE Robotics and Automation Letters* (2026). DOI: 10.1109/LRA.2026.3664615.

[4] **Huang, Yunqi** and T. G. Thuruthel. "Electrical Impedance Tomography Based Finger-Shaped Soft Artificial Skin". *IEEE Robotics and Automation Letters* (2025), pp. 1–8. DOI: 10.1109/LRA.2025.3557297.

[5] **Huang, Yunqi**, D. Hardman, C. Bascucci, F. Clemens, and T. G. Thuruthel. "Design and Modelling of Electrical Impedance Tomography-based 3D-Printed Patterned Soft Tactile skins". *2025 IEEE 8th International Conference on Soft Robotics (RoboSoft)*. 2025, pp. 1–6. DOI: 10.1109/RoboSoft63089.2025.11020951.

[6] **Huang, Yunqi**, A. Y. Alkayas, J. Shi, F. Renda, H. Wurdemann, and T. G. Thuruthel. "Predicting Interaction Shape of Soft Continuum Robots using Deep Visual Models". *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2024, pp. 11381–11387. DOI: 10.1109/IROS58592.2024.10801261.

[7] Q. Zhang, **Huang, Yunqi**, C.-B. Chng, C.-K. Chui, and P.-S. Lee. "Investigations on machine learning-based control-oriented modeling using historical thermal data of buildings". *Building and Environment* 243 (2023), p. 110595.

SERVICE

Co-organizer, Workshop on ICRA 2026 **Towards Large Area Tactile Sensing Skins**	June 2026
UCL Postgraduate Teaching Assistant (COMP0208, COMP0211, COMP0223)	2024 - Present
Journal Reviewer: T-RO, RA-L, T-BME	2024 - Present
Conference Reviewer: ICRA, IROS, RoboSoft	2023 - Present